CASE 08 · AI-AUGMENTED BUSINESS ANALYST PROJECT · EDUCATION & LEARNING ANALYTICS

The 53 Percent Case for Deeper Content

An LLM-powered qualitative theme-coding pilot on real Coursera course reviews - turning "why are learners unhappy" into a taxonomy a curriculum team can act on.

Role simulated: Learning & Development / Curriculum Analytics Analyst | Tools: Python, LLM-based open qualitative coding, thematic cross-tabulation

Source: 107,018 real Coursera platform reviews across all courses/subjects (public dataset); n=120 stratified reading sample, hand-coded by the analyst-as-LLM

120
reviews LLM-coded (of 107,018 on the platform)

39%
of the sample tagged “Shallow, thin, or outdated content”

53%
of 1-2 star reviews carry that same theme

8
themes discovered by open coding

The business problem

A learning & development / curriculum analytics team wants to know, at scale, why learners are unhappy with online courses - not just that the average rating is 4.2 stars, but actionable themes a curriculum team can act on. Manually reading and coding tens of thousands of reviews isn't feasible, so this project pilots an LLM-based qualitative coding pipeline on a representative stratified sample, the way an analyst would actually run this in 2026: the model does the first-pass reading and tagging, a human validates a sample and turns it into a roadmap.

Methodology - and its honest limits

The full dataset is heavily skewed positive: 79,173 five-star reviews (74.0% of all 107,018), against just 4,720 one- or two-star reviews (4.4% combined). A sample drawn to match that true distribution would hand a coder maybe a dozen unhappy reviews to learn from - not enough to build a usable complaint taxonomy. So the 120 reviews sampled here (random_state=42) were deliberately stratified toward the actionable, unhappy tail: 70 reviews at 1-2 stars, 25 at 3 stars, 25 at 4-5 stars - roughly 58% / 21% / 21%, a heavy oversample of the roughly 4% / 5% share 1-2 and 3-star reviews actually hold platform-wide. That is a deliberate design choice, not an attempt to represent the platform's true sentiment mix; the results below describe what unhappy learners say, not what share of all learners are unhappy.

The taxonomy of 8 themes below was built inductively - reading the sampled reviews first, before any tagging, and letting the categories emerge from what learners actually wrote, rather than forcing a pre-set list onto the text. Each review was then coded with 1-2 theme tags. This was a single coder (one LLM pass, zero re-reads) with no second-rater reliability check performed - the standard next step before trusting these proportions at production scale would be a second independent coding pass on a 20-30 review overlap sample, with inter-rater agreement (e.g. Cohen's kappa) reported, plus a larger sample once the taxonomy

is validated. Treat the percentages here as a directional read on what to investigate next, not a statistically powered estimate of the whole platform.

What learners are actually saying

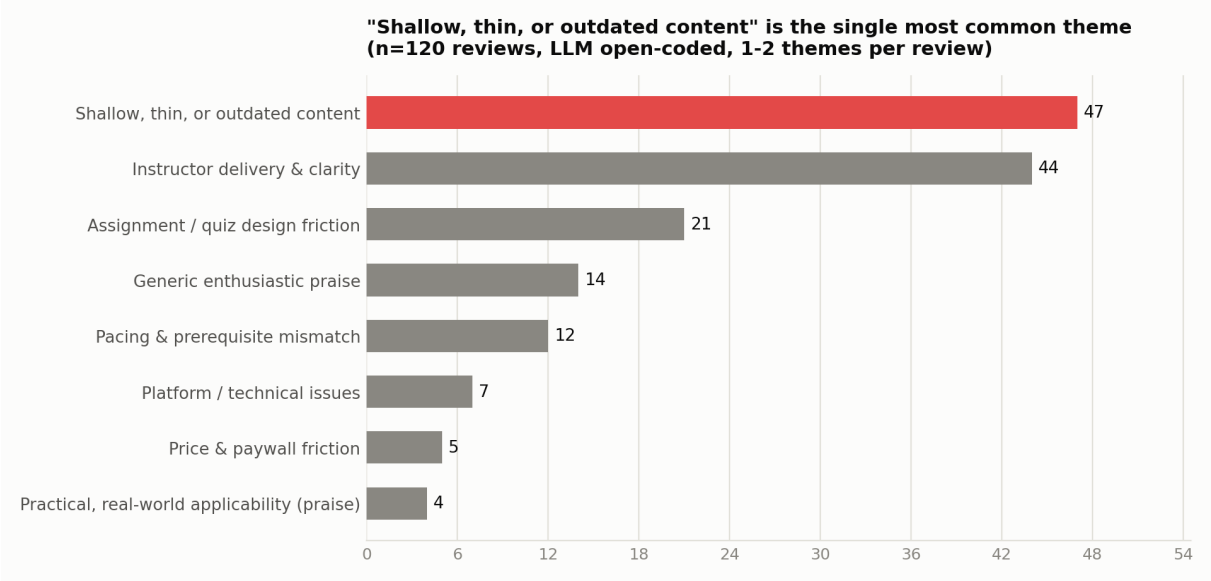


Fig. 1 - Review count per theme across the 120-review coded sample (a review can carry up to 2 themes, so counts do not sum to 120).

“Shallow, thin, or outdated content” is the single most common theme, present in 47 of 120 reviews (39%) - and it concentrates hard in the unhappy tail (below). In their own words:

“...this course is useless. It touches so many different topics without going deeply into any of them. All you get is a basic knowledge about All Things Cyber™...You can get the same amount of knowledge by reading two or three Wikipedia pages.” - 1-star review

“Only few starting videos give new information and techniques, the rest 80% is quite useless...If you read the course contents - you already get 75% of what is presented in the videos.” - 2-star review

“Unfortunately, this specific course lacks in-depth information and problems and exercises in order to get an individual to explore the wide subject...” - 1-star review

Instructor delivery & clarity is close behind at 44 reviews (37%) - but unlike content depth, this theme is double-edged: it drives both complaints (23 in the 1-2 star band) and praise (11 in the 4-5 star band). “Lousy course.....boring teacher” and “the instructor looks like she's reading the material. It's very boring to watch” sit in the same theme bucket as “the professor is very easy to follow and makes a subject that is constantly deemed as dry much more interesting.” Same theme, opposite outcome - which is exactly the signal a curriculum team needs: instructor delivery quality is not a fixed cost of a course, it is the single biggest swing factor between a 1-star and a 5-star review.

Unhappy vs. happy - the theme mix shifts

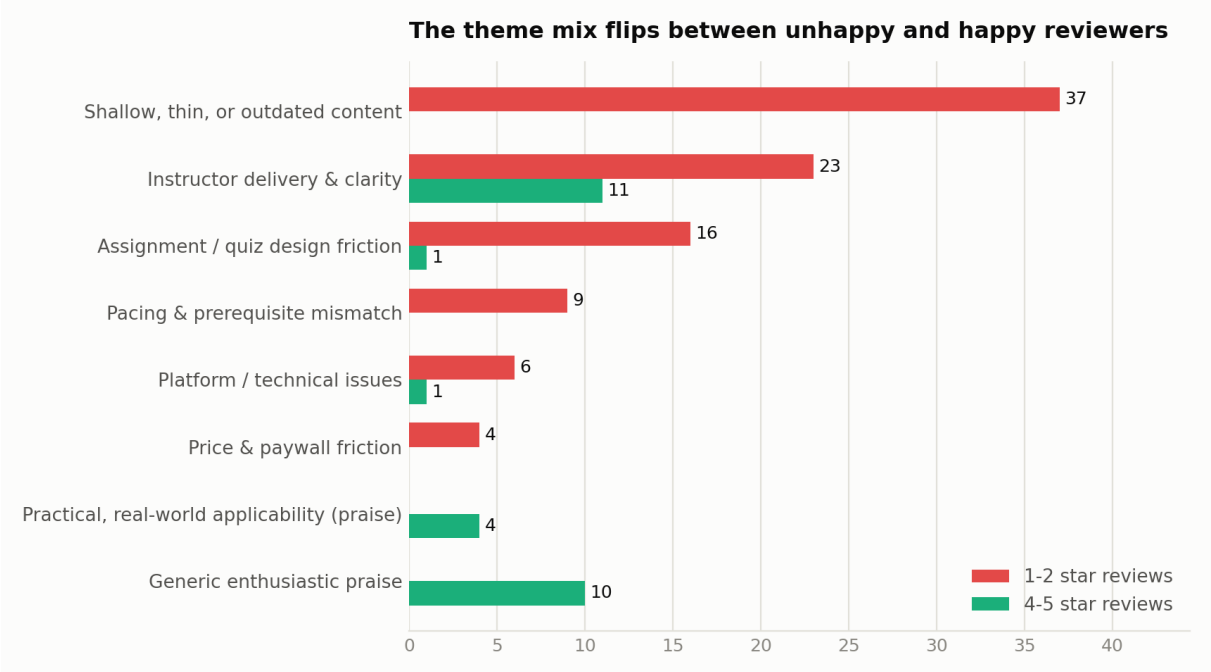


Fig. 2 - Reviews per theme, 1-2 star band vs. 4-5 star band (3-star reviews omitted from this comparison for clarity; full crosstab is in theme_mining_results.json).

“Shallow, thin, or outdated content” and “Assignment / quiz design friction” are almost entirely complaint-side themes: 37 of 70 unhappy reviews (53%) cite thin content, versus zero 4-5 star reviews; assignment friction is 16 unhappy vs. just 1 happy mention. Compare that to “Generic enthusiastic praise” and “Practical, real-world applicability,” which appear in zero unhappy reviews and cluster entirely in the happy band. Instructor delivery is the one theme with real mass on both sides, confirming it as the highest-leverage, most fixable lever available: unlike content depth (which requires a curriculum rebuild) or pricing (a business-model decision), instructor delivery quality can improve course-by-course through re-recording, pacing edits, or captioning fixes without touching the syllabus.

Recommendation

1. Prioritize a content-depth audit on the courses driving the “Shallow, thin, or outdated content” theme - the reviews describe a specific, fixable pattern: strong opening weeks that thin out badly by the back half, or breadth prioritized over depth in a way learners repeatedly compare unfavorably to “just reading a Wikipedia page.” Flag any course scoring 3+ tagged mentions in this theme for a syllabus review.

2. Treat instructor delivery as a coaching and re-recording opportunity, not a personnel problem - the same theme produces both the platform's harshest complaints and warmest praise, meaning the fix is presentation craft (pacing, energy, avoiding a read-the-slides delivery), not different instructors. A lightweight rubric (engagement, pacing, clarity) scored against a sample of lecture video would catch this before launch.

3. Route assignment/quiz design complaints to a QA pass before content review - these are mechanically fixable (misaligned lecture-to-assignment content, ambiguous peer-review prompts, quizzes testing material never taught) and represent quick wins independent of any curriculum rebuild.

RESUME BULLET, READY TO USE

“Ran an LLM-powered open qualitative coding pass on a stratified sample of 120 real Coursera reviews (of 107,018 platform-wide), inductively building an 8-theme taxonomy and cross-tabulating it against star rating to find that 53% of 1-2 star reviews cite thin/shallow content as a driver - translated into a 3-point curriculum action plan.”

What's in this package

File	What it is
run_analysis.py	Deterministic analysis: frequency table, theme x rating crosstab, and both charts - fully reproducible from the coded JSON.
llm_theme_codes.json	The actual open-coding output: all 120 reviews with star rating and the 1-2 theme tags assigned by reading each one.
This report (PDF)	Business framing, methodology and its limits, findings, and curriculum recommendations.